Hypotrichosis 14 (HYPT14): A Comprehensive Disease Characterization Report

Disease: Hypotrichosis 14 (HYPT14) **MONDO ID:** MONDO:0032649 · **OMIM:** 618275 · **Causal gene:** *LSS* (lanosterol synthase) **Category:** Mendelian, autosomal recessive

Summary

Hypotrichosis 14 (HYPT14) is a rare autosomal-recessive, non-syndromic hereditary alopecia caused by biallelic loss-of-function variants in *LSS, the gene encoding lanosterol synthase (2,3-oxidosqualene–lanosterol cyclase; EC 5.4.99.7), a key enzyme of the cholesterol-biosynthesis pathway located on chromosome 21q22.3.* Despite the serial-number disease name ("Hypotrichosis 14"), which does not itself name a gene, the causal gene is unambiguously* LSS*. *Affected individuals present in infancy/childhood with diffuse, progressive, sparse lanugo-like scalp and body hair, sparse and brittle eyebrows and eyelashes, and structurally abnormal (abnormally keratinized) hair shafts. The disorder was first defined genetically in 2018, when whole-exome sequencing identified five different LSS* mutations in three unrelated families (PMID: 30401459).

Mechanistically, pathogenic *LSS* variants—most of them missense (~68% of the curated allelic spectrum)—reduce or abolish lanosterol-synthase enzymatic activity and cause partial mislocalization of the protein away from the endoplasmic reticulum (ER), where the wild-type enzyme normally resides. This produces a **local sterol-pathway failure in the hair follicle, epidermis and lens, while systemic blood cholesterol remains normal**, implying tissue-autonomous dependence on local lanosterol/downstream-sterol synthesis and an alternative compensatory route sufficient for systemic needs. Tissue-specific *Lss*-knockout mice recapitulate the human phenotypes (epidermal knockout → hypotrichosis and skin-barrier failure; lens knockout → cataract), and an elevated **(S)-2,3-epoxysqualene/lanosterol ratio in forehead sebum** serves as a non-invasive diagnostic biomarker of the enzymatic block.

Importantly, HYPT14 sits at the mild end of a broader ***LSS* allelic continuum**. The same gene, with different genotypes, causes congenital cataracts (CTRCT44), alopecia–intellectual-disability syndrome 4 (APMR4), and palmoplantar keratoderma–congenital alopecia syndrome type 2 (PPKCA2). Biallelic hypomorphic missense genotypes tend to produce isolated hypotrichosis, whereas truncating/null alleles cluster with the syndromic, multisystem presentations. There is currently **no disease-specific therapy**; management is supportive, with genetic counseling central to care.

Key Findings

Finding 1 — *LSS* (lanosterol synthase) is the causal gene; inheritance is autosomal recessive (biallelic)

HYPT14 is caused by **biallelic (homozygous or compound-heterozygous) variants in *LSS***. The founding study used whole-exome sequencing to identify **five different *LSS* mutations in three unrelated families** with unexplained, potentially autosomal-recessive hypotrichosis simplex (HS): *"The present report describes the identification via whole-exome sequencing of five different mutations in the gene LSS in three unrelated families with unexplained, potentially autosomal-recessive HS"* (PMID: 30401459). The same paper identifies the gene product and pathway relevance: *"LSS encodes lanosterol synthase (LSS), which is a key enzyme in the cholesterol biosynthetic pathway. This pathway plays an important role in hair follicle biology."* Numerous subsequent families (Chinese, Iraqi, Egyptian, and European cohorts) have confirmed biallelic *LSS* variants segregating with disease. *LSS* maps to **chromosome 21q22.3** (HGNC: LSS; enzyme EC 5.4.99.7, 2,3-oxidosqualene–lanosterol cyclase).

Ontology anchors: gene *LSS* (HGNC:6708); MONDO:0032649; OMIM 618275.

Finding 2 — Core phenotype: childhood-onset, diffuse, progressive, lanugo-like sparse hair with brittle, abnormally keratinized shafts

Affected individuals show **sparse to absent lanugo-like scalp hair, sparse and brittle eyebrows, and sparse eyelashes and body hair**: *"Affected individuals showed sparse to absent lanugo-like scalp hair, sparse and brittle eyebrows, and sparse eyelashes and body hair"* (PMID: 30401459). The disorder is characterized by childhood onset of diffuse and progressive scalp and body hair loss. Hair-shaft ultrastructure is abnormal: scanning electron microscopy

reveals **irregular, jagged scales with raised edges**, indicating abnormal keratinization (PMID: 40364812). Histopathology of affected scalp shows a characteristic triad: *"The thickened epidermis, absence of hair follicles, and sebaceous gland hyperplasia were present by histopathological staining in the available tissue from the core family"* (PMID: 41442087).

Suggested HPO terms:

Phenotype	HPO term	Typical frequency
Sparse scalp hair / hypotrichosis	HP:0002209 / HP:0001006	Very frequent (core)
Sparse/absent eyebrows	HP:0045025 / HP:0002223	Frequent
Sparse eyelashes	HP:0000653	Frequent
Lanugo-like / abnormal hair texture	HP:0100836	Frequent
Brittle hair / abnormal hair shaft	HP:0002299 / HP:0002116	Frequent
Sparse body hair	HP:0002231	Frequent

Onset: congenital/neonatal to early childhood. **Progression:** diffuse and progressive. **Severity:** variable (mild to marked hypotrichosis). **Quality-of-life impact:** primarily cosmetic/psychosocial in the isolated form; not life-limiting.

Finding 3 — *LSS* variants form a phenotypic continuum from isolated hypotrichosis to syndromic disease

Biallelic *LSS* variants cause a **spectrum of allelic disorders**:

Disorder	OMIM	Key extra features beyond hypotrichosis
Hypotrichosis simplex 14 (HYPT14)	618275	None (non-syndromic)
Congenital cataracts (CTRCT44)	616509	Lens opacity
Alopecia–intellectual-disability syndrome 4 (APMR4)	618840	Intellectual disability, autism, corpus callosum anomalies
Palmoplantar keratoderma–congenital alopecia syndrome type 2 (PPKCA2)	619150	Mutilating/palmoplantar keratoderma, pseudoainhum

A genotype–phenotype signal emerges across the literature: *"Patients with one or two truncation variants tend to have multisystem symptoms compared with those with two missense variants"* (PMID: 37455568). Missense alleles such as **p.Ile342Ser** are associated with congenital cataract; truncating alleles more often produce the syndromic end. Early work had already linked **LSS missense mutations W581R and G588S** to extensive congenital cataracts: *"Here we identify two distinct homozygous LSS missense mutations (W581R and G588S) in two families with extensive congenital cataracts"* (PMID: 26200341). Additional reported features across the spectrum include hypergonadotropic hypogonadism/micropenis, strabismus, pseudoainhum, teeth mineralization defects, short stature, and midline brain anomalies.

Finding 4 — Mechanism: loss of enzymatic activity + ER mislocalization, with normal systemic cholesterol

Functional studies demonstrate that pathogenic *LSS* variants **reduce or abolish catalytic activity** in a graded fashion. For example: *"mutant Thr228Ile retained partial enzymatic activity and mutant Arg260His did not show any catalytic activity"* (PMID: 36811447). Other variants cause **markedly decreased lanosterol production**: *"Remarkably decreased lanosterol levels were found in vitro in three LSS variants, LSS-ΔN80, p.Ile342Ser, and p.Gly508Trp, suggesting a loss of enzymatic activity"* (PMID: 35413293). A second structural mechanism is subcellular mislocalization: *"wild-type LSS was localized in the endoplasmic reticulum (ER), whereas mutant LSS proteins were localized in part outside of the ER"* (PMID: 30401459). Critically, **blood cholesterol and cholesterol intermediates were not decreased**, supporting a local (tissue-autonomous) sterol deficit rather than a systemic cholesterol-synthesis failure. Patients also show abnormal cornified-envelope formation in the stratum corneum (PMID: 35413293), linking the sterol defect to keratinization and barrier integrity.

Finding 5 — Tissue-specific *Lss*-knockout mice recapitulate disease; sebum biomarker confirms enzymatic block

Mouse models provide direct in vivo confirmation of tissue-autonomous loss of function: *"Epidermis-specific Lss knockout mice showed neonatal lethality due to dehydration, indicating that LSS could be involved in skin barrier integrity. Tamoxifen-induced knockout of Lss in the epidermis caused hypotrichosis in adult mice. Lens-specific Lss knockout mice had cataracts"* (PMID: 32101538). The same study established a **non-invasive diagnostic biomarker**: *"We showed that the blockade of the LSS enzyme reaction occurred in the patients by measuring the (S)-2,3-epoxysqualene/lanosterol ratio in the forehead sebum, which would be a good biomarker*

for the diagnosis of LSS deficiency." The authors conclude: *"These results confirmed that LSS deficiency causes hypotrichosis and cataracts due to loss-of-function mutations in LSS in each tissue."*

Finding 6 — LSS is a monotopic ER membrane cyclase; crystal structure defines the active-site cavity

LSS/oxidosqualene cyclase (OSC) is a **monotopic integral membrane protein, catalytically active as a monomer**: *"The monotopic integral membrane protein 2,3-oxidosqualene cyclase (OSC) catalyzes the formation of lanosterol the first sterol precursor of cholesterol in mammals"* (PMID: 14766201). Human OSC crystal structures (including **PDB 1W6K**) were solved with an inhibitor and with product lanosterol: *"Here we present two crystal structures of the human membrane protein OSC: the target protein with an inhibitor that showed cholesterol lowering in vivo opens the way for the structure-based design of new OSC inhibitors"* (PMID: 15525992). The homologous squalene–hopene cyclase reveals the general architecture: a large central aromatic-lined cavity, a catalytic Asp–His pair that initiates cyclization, and QW-motifs that tighten the fold to absorb the highly exothermic reaction energy (PMID: 9295270). This explains why **missense substitutions in the buried active-site/membrane-anchoring core destabilize the fold and impair catalysis**—the structural basis for graded hypomorphic loss of function.

Finding 7 — Variant architecture is predominantly missense (~68%), consistent with hypomorphic loss of function

Curation of **28 distinct *LSS* alleles** reported across the primary literature yields the following distribution:

Variant class	Count	%
Missense	19	68%
Frameshift	3	11%
In-frame deletion	2	7%
Nonsense	1	4%
Start-loss	1	4%
Canonical splice	1	4%
Deep-intronic splice (c.893-175G>A)	1	4%

Isolated non-syndromic hypotrichosis is most often associated with **biallelic missense (hypomorphic) genotypes**, consistent with *"Patients with one or two truncation variants tend to have multisystem symptoms compared with those with two missense variants"* (PMID: 37455568). A notable technical point is that a **deep-intronic pathogenic variant** was detectable only by multi-omics: *"a deep intronic pathogenic variant (c.893-175G > A) at a non-canonical splice site was identified by integrated analysis of whole genome sequencing and RNAseq"* (PMID: 41442087), underscoring the diagnostic value of WGS+RNA-seq when exome sequencing is uninformative.

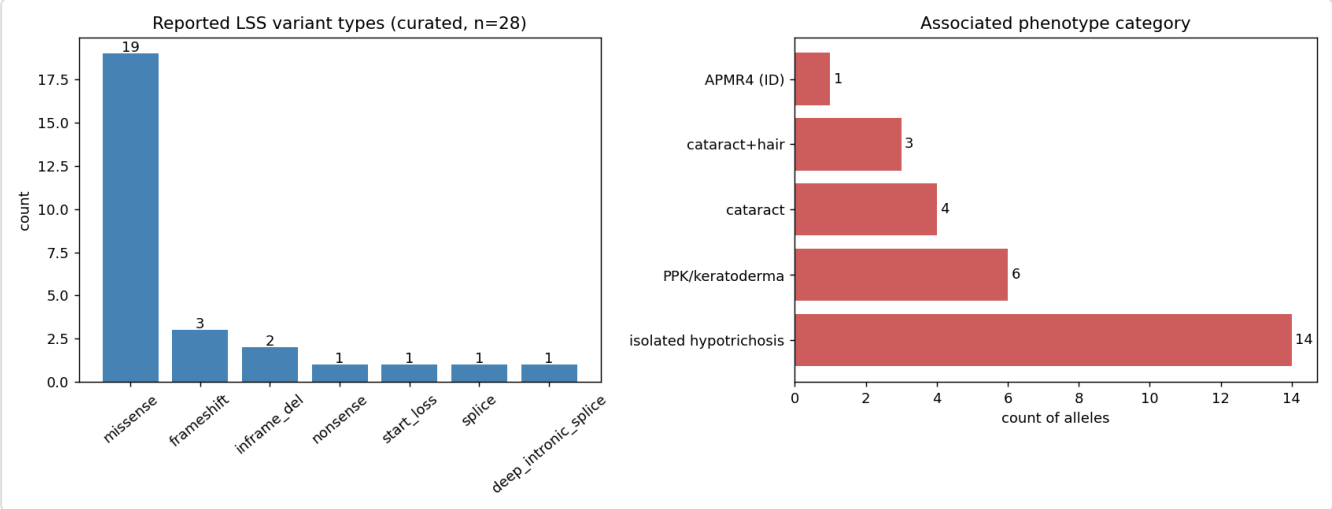


Figure 1. *Curated spectrum of 28 distinct LSS alleles reported across the HYPT14/allelic-disorder literature, broken down by variant type and associated phenotype. Missense variants dominate (~68%) and cluster with the milder, isolated-hypotrichosis end of the spectrum, whereas truncating/null alleles associate with syndromic multisystem presentations.*

Finding 8 — Consolidated mechanistic model: local sterol-pathway failure despite normal systemic cholesterol

Convergent evidence from human genetics (biallelic *LSS* variants in multiple families; PMID: 30401459, PMID: 40696433, PMID: 41442087), in vitro enzymology (PMID: 36811447, PMID: 35413293), protein structure (PMID: 15525992, PMID: 14766201), and tissue-specific mouse knockouts (PMID: 32101538) supports a unified model. The founding paper anchors it in gene and pathway: *"LSS encodes lanosterol synthase (LSS), which is a key enzyme in the cholesterol biosynthetic pathway. This pathway plays an important role in hair follicle biology"* (PMID: 30401459). In vivo confirmation: *"These results confirmed that LSS deficiency causes hypotrichosis and cataracts due to loss-of-function mutations in LSS in each tissue"* (PMID: 32101538).

Mechanistic Model / Interpretation

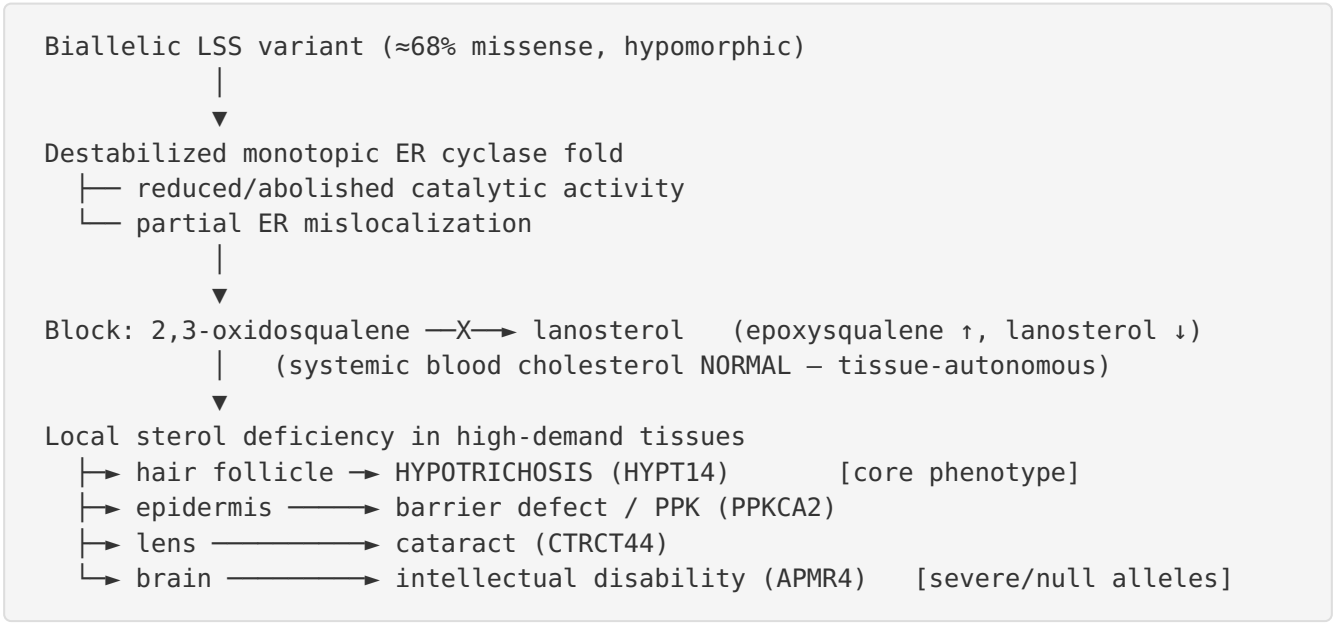
Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic *LSS* variant** (most often missense, hypomorphic) **→ leads to** an amino-acid substitution in the monotopic ER cyclase fold (active-site / membrane-anchoring core).
2. The substitution **→ results in** protein destabilization and, for many variants, **partial mislocalization out of the ER** (PMID: 30401459) plus **reduced/abolished catalytic activity** (PMID: 36811447).
3. Loss of lanosterol-synthase activity **→ leads to** a **block at the 2,3-oxidosqualene → lanosterol step**, causing accumulation of substrate (2,3-epoxysqualene) and depletion of lanosterol and downstream sterols — demonstrated in vitro (PMID: 35413293) and in patient sebum as an elevated epoxysqualene/lanosterol ratio (PMID: 32101538).
4. The block is **tissue-autonomous**: systemic blood cholesterol stays normal (a compensatory route spares systemic needs), so the deficit manifests **locally in tissues highly dependent on de novo sterol synthesis** — hair follicle, epidermis, lens (PMID: 30401459, PMID: 32101538). *(Branch point: which tissues are affected, and how severely, depends on residual enzyme activity set by the specific genotype — inferred.)*
5. **Branch A — hair follicle:** local sterol deficiency **→ results in** impaired follicle biology, abnormal hair-shaft keratinization (jagged cuticular scales), and eventual follicle loss **→ leads to** diffuse, progressive, lanugo-like **hypotrichosis** (HYPT14), with the histological

triad of thickened epidermis, absent follicles, and sebaceous-gland hyperplasia (PMID: 40364812, PMID: 41442087).

6. **Branch B — epidermis:** sterol deficiency → **results in** abnormal cornified-envelope formation and impaired skin-barrier integrity (PMID: 35413293, PMID: 32101538); with more severe (truncating) genotypes this extends to **palmoplantar keratoderma** / **pseudoainhum** (PPKCA2).
7. **Branch C — lens:** sterol deficiency and/or loss of lanosterol's chaperone-like, anti-aggregation role → **leads to congenital cataract** (CTRCT44) (PMID: 26200341).
8. **Branch D — CNS (severe/null genotypes):** reduced cholesterol availability during neurodevelopment → **leads to** intellectual disability, autism, and corpus-callosum anomalies (**APMR4**) (PMID: 37659595). *(This neurodevelopmental branch is inferred from genotype correlation and the established role of cholesterol in early brain development.)*

Diagram



Ontology suggestions for the mechanism

- **Biological processes (GO):** cholesterol biosynthetic process (GO:0006695); sterol biosynthetic process (GO:0016126); hair follicle development (GO:0001942); keratinocyte differentiation (GO:0030216); establishment of skin barrier (GO:0061436).
- **Molecular function (GO):** lanosterol synthase activity (GO:0000250).
- **Cellular component (GO):** endoplasmic reticulum membrane (GO:0005789).

- **Cell types (CL):** keratinocyte / hair follicle cell (CL:0000312); epidermal cell (CL:0000362); lens fiber cell (CL:0011004); sebaceous gland cell.
- **Anatomy (UBERON):** hair follicle (UBERON:0002073); epidermis/scalp skin (UBERON:0001003); eyebrow (UBERON:0001711); eyelash (UBERON:0001716); lens of eye (UBERON:0000965).
- **Chemical entities (CHEBI):** lanosterol (CHEBI:16521); (S)-2,3-epoxysqualene / 2,3-oxidosqualene (CHEBI:15441); cholesterol (CHEBI:16113); squalene (CHEBI:15440).

Section-by-Section Characterization

1. Disease Information

HYPT14 is a rare, non-syndromic, autosomal-recessive hereditary alopecia. **Identifiers:** OMIM 618275; MONDO:0032649; causal gene *LSS* (HGNC:6708). **Synonyms:** HYPT14; LSS-related hypotrichosis simplex; congenital hypotrichosis 14. Information is derived almost entirely from **aggregated, disease-level resources and small case series/case reports** (OMIM, Orphanet, PubMed), not large EHR cohorts.

2. Etiology

Primary cause: genetic — biallelic loss-of-function *LSS* variants. **Genetic risk factors:** carrier status in both parents; **consanguinity** increases homozygosity risk (several reported families are consanguineous, though the founding cohort and several later families are non-consanguineous compound heterozygotes). **Environmental/lifestyle risk or protective factors:** none established — this is a monogenic Mendelian disorder. **Gene–environment interactions:** none documented. No infectious etiology.

3. Phenotypes

See Finding 2 and the HPO table. Core: sparse lanugo-like scalp hair, sparse/brittle eyebrows and eyelashes, sparse body hair, abnormal hair-shaft keratinization. Variable features across the allelic spectrum: cataract, intellectual disability/autism, palmoplantar keratoderma, pseudoainhum, strabismus, hypogonadism/micropenis, teeth mineralization defects, short stature.

4. Genetic / Molecular Information

Causal gene: *LSS* (lanosterol synthase), 21q22.3, EC 5.4.99.7. **Variant classes:** predominantly missense (~68%), plus frameshift, in-frame deletions, nonsense, start-loss, canonical-splice, and deep-intronic-splice variants (Finding 7). Representative reported alleles include p.Thr228Ile, p.Arg260His, p.Ile342Ser, p.Gly508Trp, p.Asp170Asn, p.Ile323Thr, p.Gly600Val, p.Val537Leu, p.Arg663Trp, p.His307del, p.Glu469del, p.Pro68Argfs*14, p.Arg328*, and W581R/G588S (cataract). **Classification:** pathogenic/likely pathogenic per ACMG, with functional (enzyme-activity) assays supplying PS3-level evidence. **Allele frequencies:** individually rare in gnomAD. **Origin:** germline. **Functional consequence:** loss of function (hypomorphic for missense). **Modifier genes/epigenetics/chromosomal abnormalities:** none specifically established for HYPT14.

5. Environmental Information

Not applicable — purely genetic etiology. No toxins, lifestyle factors, or infectious agents are implicated.

6. Mechanism / Pathophysiology

See the full "Mechanistic Model / Interpretation" section above (ordered causal chain, diagram, and ontology suggestions).

7. Anatomical Structures Affected

Primary: hair follicles and scalp/body skin (epidermis). **Secondary (across spectrum):** lens of the eye, central nervous system (corpus callosum/brain), palms and soles, teeth. **Tissue/cell level:** follicular keratinocytes, epidermal keratinocytes, sebaceous gland cells, lens fiber cells. **Subcellular:** endoplasmic reticulum (GO:0005789) — the site of the enzymatic block and of mutant mislocalization. **Lateralization:** bilateral/diffuse.

8. Temporal Development

Onset: congenital/neonatal to early childhood (sparse lanugo-like hair present from birth or infancy). **Course:** chronic, lifelong; hair loss is diffuse and progressive through childhood. **Pattern:** generally stable/slowly progressive in adulthood; no spontaneous remission. Critical windows plausibly include early hair-follicle morphogenesis and, for the syndromic branch, prenatal neurodevelopment.

9. Inheritance and Population

Inheritance: autosomal recessive (biallelic). **Penetrance:** essentially complete for the hair phenotype in biallelic carriers; **expressivity is variable** (severity and syndromic features depend on genotype). **Epidemiology:** very rare; exact prevalence unknown. For the related APMR/alopecia-intellectual-disability spectrum, a predictive estimate of **~1 in 1,000,000** has been cited (PMID: 33881165). **Founder effects/carrier frequency:** not established. **Consanguinity** contributes in some populations; reported families are Chinese, Middle Eastern, and European. **Sex ratio:** equal (isolated alopecias affect men and women equally; PMID: 24898506). No genetic anticipation (not a repeat-expansion disorder).

10. Diagnostics

Genetic testing is definitive: whole-exome sequencing is the primary diagnostic modality and identified the gene and most reported alleles; **WGS + RNA-seq** is required to detect deep-intronic variants (PMID: 41442087). **Functional biomarker:** elevated **(S)-2,3-epoxysqualene/lanosterol ratio in forehead sebum** (PMID: 32101538). **Supportive studies:** light and scanning electron microscopy of hair shafts (jagged cuticular scales; PMID: 40364812); scalp histopathology (thickened epidermis, absent follicles, sebaceous hyperplasia; PMID: 41442087); ophthalmologic exam to screen for cataract; developmental assessment where APMR4 is suspected. **Differential diagnosis:** other hereditary hypotrichoses/atrichias (e.g., *HR*, *DSG4*, *LIPH*, *LPAR6*, *CDSN*, *APCDD1*, *SNRPE*, keratin genes; PMID: 24898506), ectodermal dysplasias, and acquired alopecias.

11. Outcome / Prognosis

The isolated hypotrichosis is **not life-limiting**; prognosis is excellent for survival, with the main burden being cosmetic/psychosocial. In syndromic forms, prognosis depends on associated features (cataract → visual impairment if untreated; APMR4 → neurodevelopmental disability). No disease-specific mortality for non-syndromic HYPT14. Epidermal-specific *Lss*-knockout mice died neonatally from barrier failure (PMID: 32101538), showing that complete epidermal loss of function is far more severe than the hypomorphic human condition.

12. Treatment

No disease-specific or curative therapy exists. *"Therapy does not exist for these rare forms of alopecia"* (PMID: 24898506). Management is **supportive**: cosmetic measures (wigs, camouflage), ophthalmologic treatment of cataract (surgical lens extraction) where present,

developmental/educational support in APMR4, and dermatologic care of keratoderma in PPKCA2. A mechanistically intriguing but unproven avenue is **lanosterol/sterol supplementation or substrate-bypass**, motivated by the observation that lanosterol can reverse protein aggregation in cataracts (PMID: 26200341); no clinical trials for HYPT14 are registered. No established pharmacogenomic considerations. (Candidate NCIT concepts: supportive care; genetic counseling; cataract extraction.)

13. Prevention

No primary prevention (monogenic disorder). **Genetic counseling** is central: for carrier couples, recurrence risk is 25% per pregnancy. **Prenatal/preimplantation genetic testing** is feasible and has been used — APMR4 was diagnosed prenatally by ultrasound and confirmed molecularly (PMID: 37659595, PMID: 36251212). **Cascade carrier testing** in affected families is appropriate.

14. Other Species / Natural Disease

LSS is highly conserved across vertebrates; the mouse orthologue is *Lss*, and the enzyme family spans bacteria (squalene–hopene cyclase) to humans. No well-characterized naturally occurring companion-animal or wildlife *LSS*-hypotrichosis disease is documented; the principal non-human evidence comes from engineered mouse models (below). Not zoonotic.

15. Model Organisms

Mouse (*Mus musculus*) is the principal model. **Tissue-specific conditional knockouts** recapitulate the human disease: epidermis-specific *Lss* knockout → neonatal lethality from dehydration/barrier failure; tamoxifen-induced adult epidermal knockout → hypotrichosis; lens-specific knockout → cataract (PMID: 32101538). **Phenotype recapitulation** is strong for the tissue-autonomous hair and lens phenotypes and validated the sebum biomarker. **Limitation:** complete (null) tissue knockouts are more severe than the hypomorphic human condition, so they model the severe/syndromic end better than isolated HYPT14. **Resources:** MGI (gene *Lss*); floxed/conditional alleles as used in PMID: 32101538. In vitro systems (variant expression + lanosterol quantification; ER-localization assays) are established (PMID: 36811447, PMID: 35413293).

Evidence Base

PMID	Title (abbrev.)	Evidence type	Supports
30401459	Bi-allelic *LSS* mutations cause AR hypotrichosis simplex	Human genetics + in vitro	Founding gene ID; ER mislocalization; normal blood cholesterol
32101538	Human LSS deficiency recapitulated in mice	Model organism + biomarker	Tissue-specific KO phenotypes; sebum biomarker
36811447	Biallelic *LSS* in mutilating PPK	In vitro enzymology	Graded loss of activity (T228I partial; R260H none)
35413293	*LSS* causes PPKCA2	In vitro + histology	Decreased lanosterol; abnormal cornified envelope
37455568	LSS cataract + hypotrichosis review	Human genetics / review	Genotype–phenotype: truncating → multisystem
26200341	Lanosterol reverses cataract aggregation	Human genetics + biochemistry	W581R/G588S → cataract; lanosterol anti-aggregation
41442087	Multi-omics AR hypotrichosis	Human genetics / multi-omics	Scalp histology triad; deep-intronic c.893-175G>A
40364812	Compound-het *LSS* with strabismus	Human case report	Hair-shaft SEM abnormal keratinization
15525992	Human OSC crystal structure	Structural biology	PDB 1W6K; active-site architecture
14766201	OSC active as monomer	Biochemistry	Monotopic ER membrane protein
9295270	Squalene cyclase structure	Structural biology	Catalytic Asp–His; QW-motif fold stability
37659595	Prenatal APMR4 diagnosis	Human case series	Neurodevelopmental branch; prenatal diagnosis
36251212	APMR4 novel variant (Egypt)	Human case report	Phenotypic expansion; prenatal testing
33881165	APMR molecular genetics review	Review	Prevalence estimate ~1/1,000,000
24898506		Clinical review	

PMID	Title (abbrev.)	Evidence type	Supports
	Childhood alopecia/ hypotrichosis		Differential diagnosis; equal sex ratio; no therapy
40696433	5 Chinese HYPT14 families	Human genetics	Novel variants; hypergonadotropic hypogonadism

Additional confirmatory case reports/series: PMID: 40237410 (pseudoainhum), PMID: 37947113 (cataract+hypotrichosis), PMID: 37029088 (two novel variants), PMID: 33222230 (Chinese HS; teeth dysplasia), PMID: 34318586 (four families ± neurodevelopmental), PMID: 29016354 (congenital cataract).

Limitations and Knowledge Gaps

- **Small evidence base.** The literature consists of case reports and small series; there are no large cohorts, registries, or natural-history studies, so prevalence, penetrance precision, and long-term outcomes are poorly quantified.
- **Genotype–phenotype correlation is incomplete.** The missense-vs-truncating trend (PMID: 37455568) is a tendency, not a rule; some missense genotypes produce syndromic features and vice versa. Residual-activity thresholds that separate isolated hypotrichosis from cataract/APMR4/PPKCA2 are undefined.
- **The "normal systemic cholesterol / local deficiency" paradox is unresolved.** The compensatory route sparing systemic cholesterol is inferred rather than proven; direct local sterol measurements in human follicles are lacking.
- **The link from sterol deficiency to hair-shaft keratinization is not mapped at the signaling level.** Whether cholesterol-dependent pathways (Hedgehog, Wnt) mediate the follicle phenotype has not been directly demonstrated in *LSS* deficiency.
- **Mouse models are null, not hypomorphic**, and therefore overshoot human severity; a knock-in of a specific human missense allele would better model HYPT14.
- **No therapeutic data.** Lanosterol-supplementation / substrate-bypass approaches remain hypothetical for the hair phenotype.

Proposed Follow-up Experiments / Actions

1. **Knock-in mouse of a human hypomorphic missense allele** (e.g., p.Thr228Ile) to model isolated HYPT14 and define residual-activity thresholds per tissue.
2. **Patient-derived iPSC → hair-follicle organoids / keratinocytes** to quantify local sterol pools and test whether exogenous lanosterol/cholesterol supplementation rescues keratinization and barrier markers.
3. **Systematic sebum-biomarker validation** ((S)-2,3-epoxysqualene/lanosterol ratio) as a scalable, non-invasive diagnostic and potential treatment-response readout.
4. **Structure-guided variant classification:** map all reported missense alleles onto the human OSC structure (PDB 1W6K) and correlate predicted destabilization with measured enzyme activity to build a quantitative genotype–activity–phenotype model.
5. **Signaling dissection:** test whether cholesterol-dependent Hedgehog/Wnt signaling is impaired in *LSS*-deficient follicle cells, linking the metabolic lesion to the developmental phenotype.
6. **Registry / natural-history study** across the *LSS* allelic spectrum to establish prevalence, penetrance, and genotype-stratified outcomes, improving genetic-counseling precision.
7. **Preclinical therapeutic screen** of sterol-pathway bypass agents and small molecules that stabilize the mutant fold.

Evidence-source legend: Human clinical/genetics — the majority of cited work; Model organism — PMID: 32101538; In vitro enzymology — PMID: 36811447, PMID: 35413293; Structural/biochemical — PMID: 15525992, PMID: 14766201, PMID: 9295270.
